Week 5 – MATH I

Fall 2026

Practical Session exercises

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MATH I - Week 5

Practical Session exercises

Quick reference

A function f is continuous at $x = a$ if:

$$f(a) \text{ is defined}, \qquad \lim_{x \to a} f(x) \text{ exists}, \qquad \lim_{x \to a} f(x) = f(a).$$

Common types of discontinuity are:

- **Removable:** the limit exists but differs from $f(a)$, or $f(a)$ is undefined.
- **Jump:** the left- and right-hand limits are finite but unequal.
- **Infinite:** the function becomes unbounded near the point.
- **Oscillatory:** the function oscillates without approaching a single value.

Exercise 1: Checking continuity

Determine whether

$$f(x) = \begin{cases} x^2 + 1, & x < 2, \\ 5, & x = 2, \\ 3x - 1, & x > 2 \end{cases}$$

is continuous at $x = 2$.

Exercise 1: Checking continuity

Determine whether

$$f(x) = \begin{cases} x^2 + 1, & x < 2, \\ 5, & x = 2, \\ 3x - 1, & x > 2 \end{cases}$$

is continuous at $x = 2$.

Solution

The left-hand limit is

$$\lim_{x \to 2^-} f(x) = 2^2 + 1 = 5.$$

The right-hand limit is

$$\lim_{x \to 2^+} f(x) = 3(2) - 1 = 5.$$

Also,

$$f(2) = 5.$$

Therefore,

$$\lim_{x \to 2} f(x) = f(2) = 5,$$

so f is continuous at $x = 2$.

Exercise 2: Removing a discontinuity

Find the value of k that makes

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3, \\ k, & x = 3 \end{cases}$$

continuous at $x = 3$.

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Solution

For $x \neq 3$,

$f(x) = \frac{(x-3)(x+3)}{x-3} = x + 3.$

Therefore,

$\lim_{x \to 3} f(x) = 6.$

Continuity requires

$k = \lim_{x \to 3} f(x).$

Hence,

$\boxed{k = 6}$.

Exercise 3: Continuous switching law

A power converter is described by

$$P(t) = \begin{cases} 4t + 1, & t < 2, \\ at^2 - 1, & t \geq 2. \end{cases}$$

Find a so that P is continuous at $t = 2$.

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$$P(t) = \begin{cases} 4t + 1, & t < 2, \\ at^2 - 1, & t \geq 2. \end{cases}$$

Find a so that P is continuous at $t = 2$.

Solution

The left-hand limit is

$$\lim_{t \to 2^-} P(t) = 4(2) + 1 = 9.$$

From the right,

$$P(2) = 4a - 1.$$

For continuity,

$$4a - 1 = 9.$$

Thus,

$$4a = 10$$

and

$$\boxed{a = \frac{5}{2}}.$$

Exercise 4: Composition and continuity

Let

$$f(x) = \sqrt{x}, \qquad g(t) = 4 - t^2.$$

Determine the domain of

$$h(t) = f(g(t))$$

and state where h is continuous.

Exercise 4: Composition and continuity

Let

$$f(x) = \sqrt{x}, \qquad g(t) = 4 - t^2.$$

Determine the domain of

$$h(t) = f(g(t))$$

and state where h is continuous.

Solution

The composition is

$$h(t) = \sqrt{4 - t^2}.$$

The expression inside the square root must be nonnegative:

$$4 - t^2 \geq 0.$$

Therefore,

$$-2 \leq t \leq 2.$$

Both g and the square-root function are continuous on the relevant domains. Hence,

$$\boxed{h(t) = \sqrt{4 - t^2} \text{ is continuous on } [-2, 2].}$$

Exercise 5: Finding an inverse

The displacement x of an actuator is related to input voltage V by

$$x(V) = 0.5V + 2, \qquad 0 \leq V \leq 10.$$

1. Find the range of x.
2. Find the inverse $V(x)$.
3. Explain why the inverse is continuous.

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2. Find the inverse $V(x)$.
3. Explain why the inverse is continuous.

Solution

At the endpoints,

$$x(0) = 2, \qquad x(10) = 7.$$

Because $x(V)$ is increasing, its range is

$$[2, 7].$$

To find the inverse, solve

$$x = 0.5V + 2.$$

Then

$$x - 2 = 0.5V,$$

so

$$\boxed{V(x) = 2(x - 2) = 2x - 4}, \qquad 2 \le x \le 7.$$

The original function is continuous and strictly increasing on $[0, 10]$. Therefore, its inverse is continuous on $[2, 7]$.

Exercise 6: Why a restriction is necessary

Consider

$f(x) = x^2$.

1. Explain why f has no inverse on $\mathbb{R}$.
2. Find its inverse if the domain is restricted to $[0, \infty)$.
3. Find its inverse if the domain is restricted to $(-\infty, 0]$.

Exercise 6: Why a restriction is necessary

Consider

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Solution

On $\mathbb{R}$,

$f(-x) = f(x)$,

so different inputs can produce the same output. For example,

$f(-3) = f(3) = 9$.

Thus, f is not one-to-one on $\mathbb{R}$.

On $[0, \infty)$, f is strictly increasing. Therefore,

$\boxed{f^{-1}(y) = \sqrt{y}}$.

On $(-\infty, 0]$, f is strictly decreasing!

Why? Because as x gets greater values the function gets smaller values.

For example, as x increases toward zero,

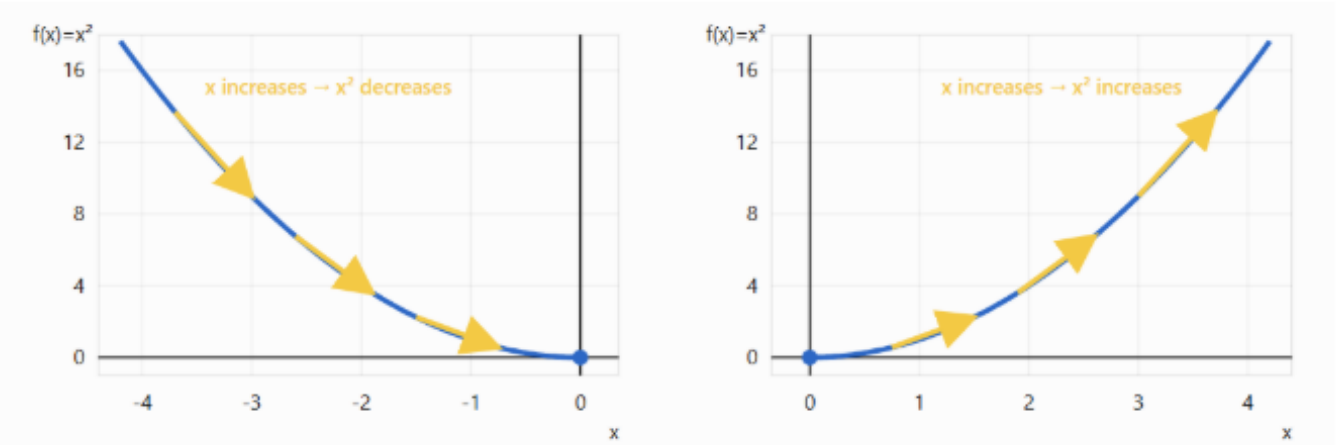
$$-4 < -3 < -2 < -1 < 0,$$

but the corresponding squares are

$$16 > 9 > 4 > 1 > 0.$$

Thus, larger inputs produce smaller outputs.

The following schematics help elucidating this: on $(-\infty, 0]$, the function f is strictly decreasing, whereas on $[0, +\infty)$, the function f is strictly increasing.



The arrows follow increasing x: x^2 decreases toward 0 on the negative branch and increases away from 0 on the positive branch.

The inverse must return a nonpositive number, so

$$\boxed{f^{-1}(y) = -\sqrt{y}}.$$

Both inverse functions are continuous for $y \geq 0$.

Exercise 7: Temperature calibration

A thermistor calibration model is

$$R(T) = 100e^{-0.02T}, \qquad T \geq 0,$$

where R is resistance in $\mathbf{k\Omega}$ and T is temperature in degrees Celsius.

1. Show that the function has an inverse.
2. Find $T(R)$.
3. Calculate the temperature when $R = 40\ \mathbf{k\Omega}$.

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1. Show that the function has an inverse.
2. Find $T(R)$.
3. Calculate the temperature when $R = 40\ \mathrm{k}\Omega$.

Solution

The function is continuous and strictly decreasing for $T \geq 0$. Therefore, it is one-to-one and has a continuous inverse.

Starting with

$$R = 100e^{-0.02T},$$

divide by 100:

$$\frac{R}{100} = e^{-0.02T}.$$

Take natural logarithms:

$$\ln\left(\frac{R}{100}\right) = -0.02T.$$

Therefore,

$$\boxed{T(R) = -50\ln\left(\frac{R}{100}\right)}.$$

For $R = 40$,

$$T = -50\ln(0.4) \approx 45.8.$$

Exercise 8: Applying the Intermediate Value Theorem

The temperature of a component is modelled by

$$T(t) = 20 + 5t - t^2, \qquad 0 \le t \le 4.$$

Show that the component reaches $25^\circ\mathrm{C}$ during this interval.

Solution

The function is a polynomial and is therefore continuous.

Evaluate it at two convenient times:

$$T(1) = 20 + 5 - 1 = 24,$$

and

$$T(2) = 20 + 10 - 4 = 26.$$

Because

$$24 < 25 < 26,$$

the Intermediate Value Theorem guarantees at least one $c \in (1, 2)$ such that

$$T(c) = 25.$$

For the exact times,

$$20 + 5t - t^2 = 25,$$

so

$$t^2 - 5t + 5 = 0.$$

Hence,

$t = \frac{5 \pm \sqrt{5}}{2}$.

Exercise 8: Applying the Intermediate Value Theorem

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$$T(t) = 20 + 5t - t^2, \qquad 0 \le t \le 4.$$

Show that the component reaches $25^\circ\mathrm{C}$ during this interval.

Both values lie in $[0, 4]$, so the component reaches $25^\circ\mathrm{C}$ twice:

$$\boxed{t = \frac{5 - \sqrt{5}}{2} \approx 1.382}$$

and

$$\boxed{t = \frac{5 + \sqrt{5}}{2} \approx 3.618}.$$

The theorem guarantees existence; solving the equation provides the actual times.

Exercise 9: Applying Weierstrass' theorem

The power output of a small device is modelled by

$$P(t) = -2t^2 + 12t + 5, \qquad 0 \leq t \leq 5.$$

1. Explain why P has an absolute maximum and minimum.
2. Find both extrema.

For part 2 (Find both extrema) wait for next week
we need derivatives!

Exercise 9: Applying Weierstrass' theorem

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$$P(t) = -2t^2 + 12t + 5, \qquad 0 \le t \le 5.$$

1. Explain why P has an absolute maximum and minimum.
2. Find both extrema.

Solution

The function is a polynomial, so it is continuous. Its domain $[0, 5]$ is closed and bounded. Therefore, Weierstrass' theorem guarantees both an absolute maximum and an absolute minimum.

Exercise 10: Why the closed interval matters

Consider the efficiency model

$$\eta(v) = v, \qquad 0 < v < 10.$$

Does Weierstrass' theorem guarantee a maximum and minimum?

Exercise 10: Why the closed interval matters

Consider the efficiency model

$$\eta(v) = v, \qquad 0 < v < 10.$$

Does Weierstrass' theorem guarantee a maximum and minimum?

Solution

The function is continuous, but the interval $(0, 10)$ is open rather than closed.

The function approaches 0 as $v \to 0^+$, but never reaches 0. It approaches 10 as $v \to 10^-$, but never reaches 10.

Therefore, the function has neither an absolute minimum nor an absolute maximum on this domain.

Weierstrass' theorem does not apply because the domain is not closed.

Exercise 11: A design-optimization problem

The temperature along a metal component is approximated by

$$T(x) = 40 + 12x - 3x^2, \qquad 0 \le x \le 5,$$

where x is measured in metres.

1. Use Weierstrass' theorem to justify the existence of hottest and coldest points.
2. Find their locations and temperatures.

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1. Use Weierstrass' theorem to justify the existence of hottest and coldest points.
2. Find their locations and temperatures.

Solution

Because $T(x)$ is a polynomial, it is continuous. The interval $[0, 5]$ is closed and bounded. Therefore, absolute extrema exist.

Exercise 12: Connecting all three ideas

A pressure sensor is modelled by

$$V(p) = \sqrt{p+1}, \qquad 0 \leq p \leq 8.$$

1. Show that V is continuous.
2. Explain why it has a continuous inverse.
3. Find the inverse.
4. Use Weierstrass' theorem to identify the minimum and maximum sensor voltage.

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3. Find the inverse.
4. Use Weierstrass' theorem to identify the minimum and maximum sensor voltage.

Solution

The function $p + 1$ is continuous, and the square-root function is continuous for nonnegative inputs. Therefore, V is continuous on $[0, 8]$.

It is also strictly increasing, so it is one-to-one. Consequently, it has a continuous inverse.

Starting with

$$V = \sqrt{p+1},$$

square both sides:

$$V^2 = p + 1.$$

Thus,

$$\boxed{p(V) = V^2 - 1}.$$

The voltage range is obtained from the endpoints:

$$V(0) = 1, \qquad V(8) = 3.$$

By Weierstrass' theorem, these extrema are attained:

$$\boxed{V_{\text{min}} = 1 \text{ at } p = 0}$$

and

$$\boxed{V_{\text{max}} = 3 \text{ at } p = 8}.$$

The inverse is defined for

$$\boxed{1 \le V \le 3}.$$

Exercise 13: When the theorem cannot be applied

A pipe-design cost is

$$C(r) = \pi r^2 + \frac{1}{r}, \qquad 0 < r \leq 5.$$

A student argues:

> "The cost is continuous, so Weierstrass' theorem guarantees a maximum."

Evaluate this statement.

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A student argues:

> "The cost is continuous, so Weierstrass' theorem guarantees a maximum."

Evaluate this statement.

Solution

The function is continuous for $r > 0$, but the domain

$$(0, 5]$$

is not closed because it does not contain $r = 0$.

As $r \to 0^+$,

$$\frac{1}{r} \to +\infty,$$

so

$$C(r) \to +\infty.$$

Therefore, the function has no absolute maximum on $(0, 5]$. It becomes arbitrarily large as r approaches zero.

The student's reasoning is wrong because continuity alone is insufficient. Weierstrass requires a closed and bounded domain.

Exercise 14: A discontinuous protection system

A protection controller has the output

$$u(T) = \begin{cases} T, & 0 \le T < 100, \\ 0, & T = 100. \end{cases}$$

1. Is the domain closed and bounded?
2. Is u continuous?
3. Does u attain an absolute maximum?
4. Why does Weierstrass' theorem not apply?

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2. Is u continuous?
3. Does u attain an absolute maximum?
4. Why does Weierstrass' theorem not apply?

Solution

The domain $[0, 100]$ is closed and bounded.

However,

$$\lim_{T \to 100^-} u(T) = 100,$$

while

$$u(100) = 0.$$

Therefore, the function is discontinuous at $T = 100$.

The function takes values arbitrarily close to 100, but never equals 100. For every $T < 100$, another value closer to 100 produces a larger output.

Thus, u has no absolute maximum.

Weierstrass' theorem does not apply because the function is not continuous on the closed interval.

This is a useful warning: even a finite operating range does not guarantee an attained worst-case value if the model contains discontinuous switching.

Main concepts/ideas to remember

- Continuity means

$$\lim_{x \to a} f(x) = f(a).$$

- Continuous functions preserve smooth input-output behaviour.
- A function must be one-to-one to possess an inverse.
- A continuous, strictly monotonic function on an interval has a continuous inverse.
- The Intermediate Value Theorem guarantees that continuous functions take every intermediate value.
- Weierstrass' theorem guarantees that a continuous function on a closed, bounded interval attains an absolute maximum and minimum.
- Weierstrass guarantees that extrema exist; derivative methods help locate them.